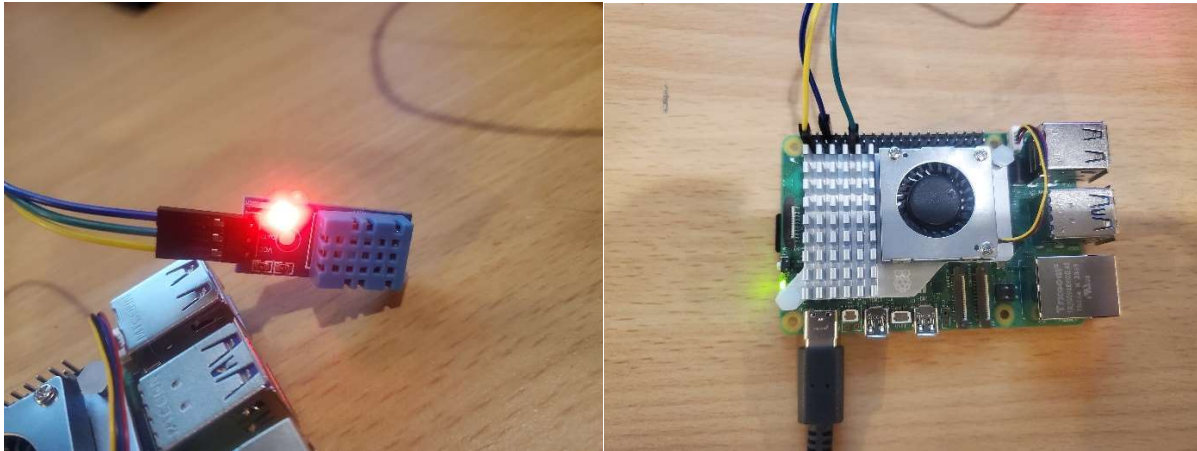


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May 3, 2024

Physical IT Systems Security Final Project Report

Project Goal

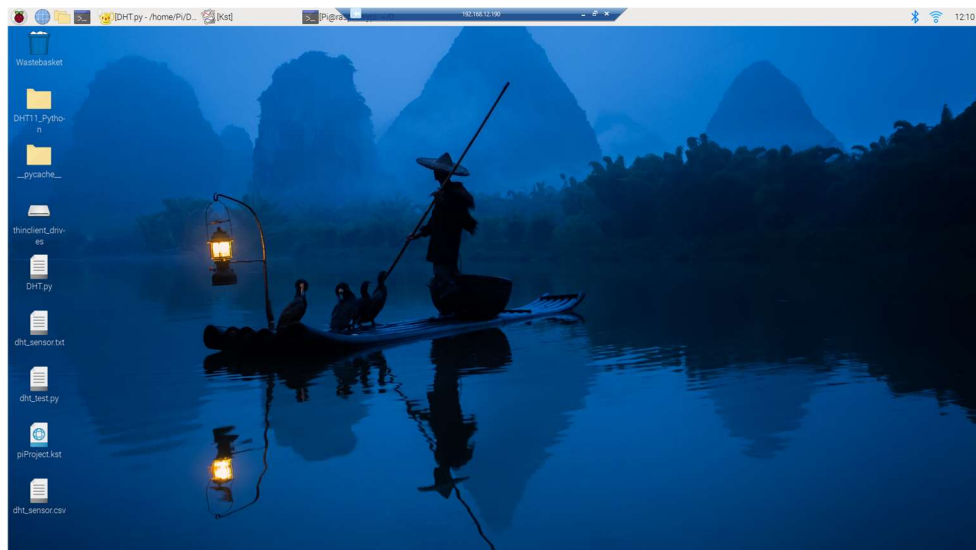
The goal of this project was to accomplish option 1 using a physical raspberry pi implement a real time data collection and visualization system. I used a raspberry Pi 5 for this project, a DHT11 Temperature and humidity sensor, and KST plot to accomplish this. Below are pictures of the sensor and the Pi.



Challenges:

Remoting in.

As mentioned in my previous report one challenge to be addressed was how I was going to remote into the pi to cut down on clutter on my desk from running the Pi. The solution I found was using RDP to remote into the Pi. I had to install and configure XRDP on the Pi and utilized the RDP software native to my windows machine to seamlessly remote in and cut down on the peripherals cluttering my desk. Using RDP, it was almost as if I had a Raspbian VM pulled up rather than my actual raspberry Pi. Below is a screenshot of my RDP window.



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Python Scripting

In my previous report I noted that I was having trouble with the example code used to operate the DHT11 sensor. I was getting an error for an “Unknown Platform” after a lot of troubleshooting and research I found out the code and python modules I was working with were not intended for a Pi 5 and were in fact deprecated after finding a tutorial intended for a Pi 5, I was able to finally get readings from the device. I also implemented a conversion for Fahrenheit for the sensor due to the sensor only collecting in Celsius. I implemented a timestamp and date collection field for the collected data entry. I also implemented a conversion to Kelvin just for fun as well.

Below is the link to the tutorial, it was tremendously helpful.

<https://www.linkedin.com/pulse/how-read-temperature-your-raspberry-pi-5-jagrat-rao-wgpoc>

Now that I could get output from the sensor, I changed focus to being able to output the data to a file. Using the python csv module the sensor data is appended to a csv file. The challenge that I found here was that the loop was only appending the sensor data when I terminated the script after a lot of research, I found that `file.flush()` could fix this problem allowing for the appended data to be written live to the CSV allowing KST to provide the real time visualization needed for this project. On the next page is the script used for this project.

Conclusion

This was definitely a fun and rewarding lab to do, I definitely see myself doing some more projects with the Pi moving forward. As requested, submitted with this report is a video demonstration of the project in action.

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Python Script

```
"""
Raspberry Pi Final Project
Author: Tomas King
This code interfaces with the Raspberry Pi and utilizes a DHT11 to take
temperature and humidity readings there is conversion done for celsius to fahrenheit
and Kelvin just for fun.
The provided documentation for this sensor was outdated and this would have been impossible
without this helpful tutorial.
https://www.linkedin.com/pulse/how-read-temperature-your-raspberry-pi-5-jagrat-rao-wgpoc
"""
import time
import board
import adafruit_dht
from datetime import datetime
import csv

# DHT11 Sensor after D is GPIO pin number
dhtDevice = adafruit_dht.DHT11(board.D17)

# Open CSV file in append 'a' mode
with open('dht_sensor.csv', 'a', encoding='UTF8', newline='') as f:
    writer = csv.writer(f)
    while True:
        try:
            # timestamp for each entry
            timestamp = datetime.now().strftime('%H:%M:%S')
            # Date for each entry
            date = datetime.now().strftime('%Y-%m-%d')
            # Actual reading from the device are in celsius
            temp_c = dhtDevice.temperature
            # Some quick math to convert to fahrenheit
            temp_f = temp_c * (9 / 5) + 32
            # Conversion to Kelvin just for fun
            temp_k = temp_c + 273.15
            # Retrieve humidity reading from sensor
            humidity = dhtDevice.humidity
            # Write data to the file
            writer.writerow([timestamp, date, temp_f, temp_c, temp_k, humidity])
            # Bypass buffer and write to the CSV immediately rather than when the script is terminated
            f.flush()
            #Print to the console whats being written to the CSV
            print(
                "Timestamp: {} {} Temp: {:.1f} F / {:.1f} C / {:.1f} K   Humidity: {}".format(
                    timestamp, date, temp_f, temp_c, temp_k, humidity
                )
            )
        except RuntimeError as error:
            print(error.args[0])
            time.sleep(2.0)
            continue
        except Exception as error:
            dhtDevice.exit()
            raise error
        # Wait for a while before the next reading
        time.sleep(2.0)
```